

A Simulation Approach Toward a Definition of Catalytic Power in Quantum Entanglement-Assisted Local Transformations

Motivation

Rapid advancements in quantum technologies have brought attention to quantum entanglement as a useful physical resource for information processing, particularly in settings where separated parties must manipulate a shared system to reach a specific outcome. This project is a computational investigation to quantify the effect of using 'borrowed' entangled states to enable conversions between two entangled states, across a range of dimensions.

Theoretical Framework

A pure bipartite quantum state $|\psi\rangle_{AB}$ is shared between two subsystems, A and B (conventionally called 'Alice' and 'Bob'). Any such state can be written in its Schmidt decomposition as:

$$|\psi\rangle_{AB} = \sum_{i=1}^n \sqrt{\lambda_i} |i\rangle_A |i\rangle_B$$

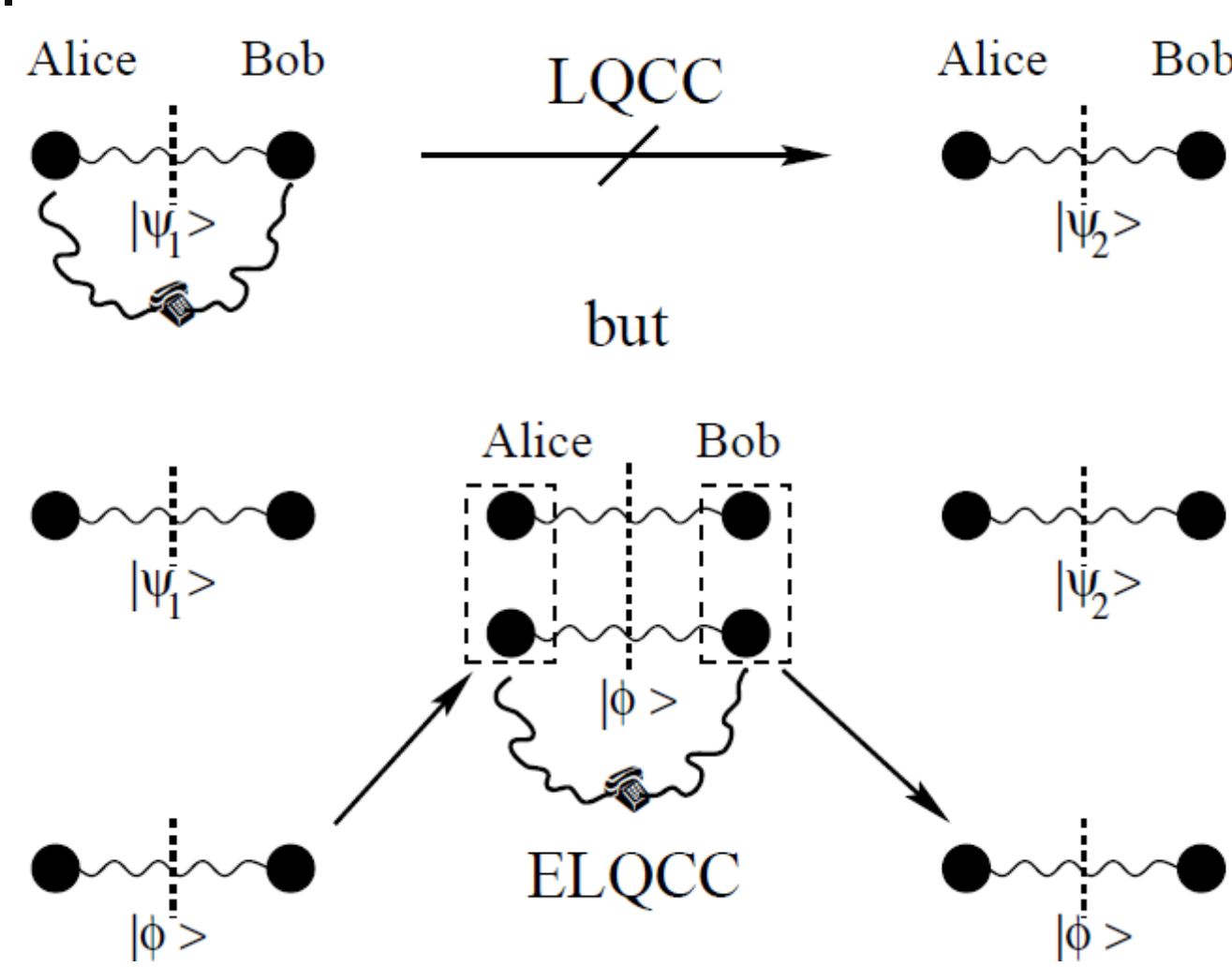
Nielsen's theorem states exactly when A and B can convert their shared state $|\psi\rangle_1$ into another $|\psi\rangle_2$ with certainty using only an LOCC protocol: when the source state's coefficient vector is majorized by the target's, that is,

$$\vec{\lambda} < \vec{\lambda}' \Leftrightarrow \sum_{i=1}^k \lambda_i \leq \sum_{i=1}^k \lambda'_i \text{ for every } k$$

However, even when a direct LOCC conversion is forbidden, borrowing an extra entangled state $|\phi\rangle$ from system C can make the joint conversion possible:

$$|\psi_1\rangle|\phi\rangle \xrightarrow{\text{LOCC}} |\psi_2\rangle|\phi\rangle \Leftrightarrow (\vec{\lambda} \otimes \vec{\beta}) < (\vec{\lambda}' \otimes \vec{\beta})$$

This auxiliary state $|\phi\rangle$ acts exactly like a catalyst in a chemical reaction: it enables an otherwise-forbidden transition without being used up, for it can be returned to C unchanged after the transformation.



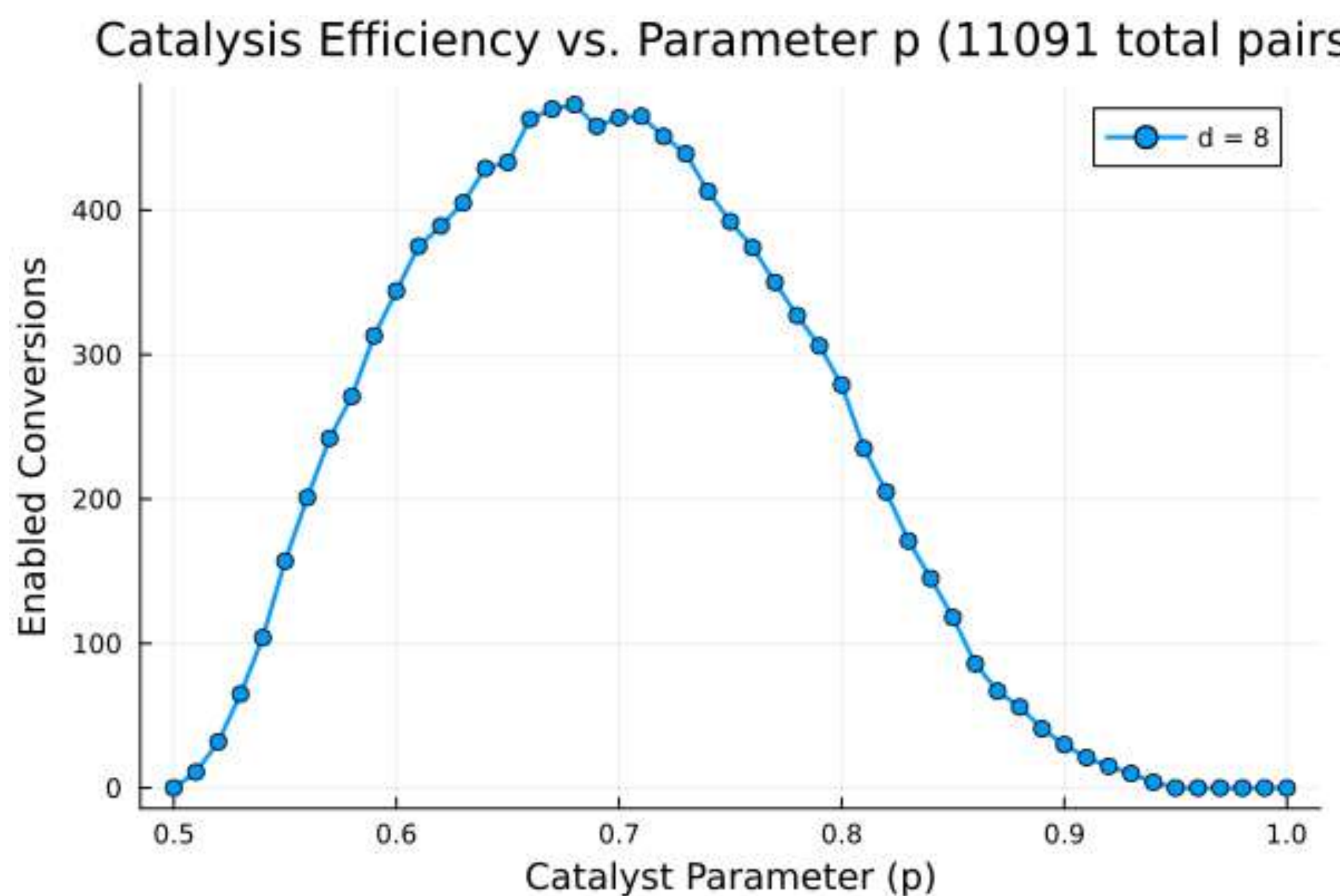
Methodology

- Sampled datasets of probability vectors (to represent bipartite states) with 20,000 and 40,000 samples drawn uniformly from a Dirichlet distribution (to avoid sampling biases).
- Split datasets into two disjoint halves of equal size, and paired states independently between them (to preserve statistical validity).
- For each incomparable pair of states, tested whether a candidate catalyst makes the joint conversion possible under Nielsen's theorem.
- Ran the catalysis-testing algorithm over two sets of candidate catalysts: one spanning two dimensions, the other three.
- Measured catalytic power as the fraction of otherwise-impossible conversions a given catalyst successfully enables.

Results for 2-dimensional catalyst

- $\vec{\beta} = (p, 1-p)$, where $p \in [0,1]$.
- The optimal catalyst always avoids the extremes of maximum and minimum entanglement.

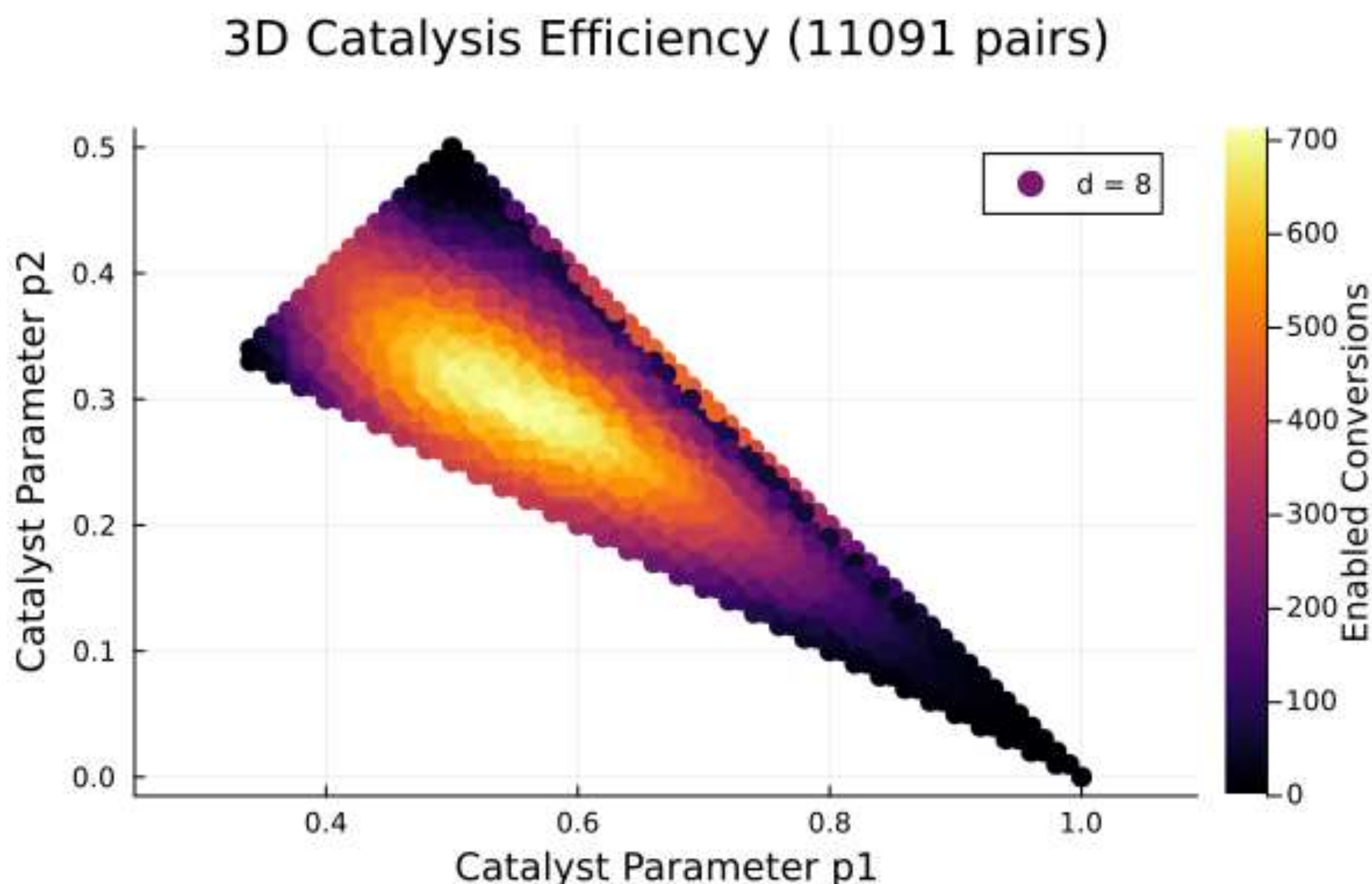
	d = 4	d = 8	d = 12
Best p	0.65	0.68	0.70
Incomparable pairs	37%	55%	64%
Catalyzing ratio	2.4%	4.3%	4.8%



Results for 3-dimensional catalyst

- $\vec{\beta} = (a, b, 1-a-b)$, where $a \in [0,1]$ and $b \in [0,1-a]$
- The optimal catalyst always lies inside the simplex, never at a corner or edge.
- The coordinates of the optimal catalyst shift with dimension — there is no single, fixed optimum.

	d = 4	d = 8	d = 12
Catalyzing ratio	3.27%	6.43%	7.26%
$\vec{\beta} = (a, b, \sim 0)$	2.37%	4.26%	4.84%
$\vec{\beta} = (a, \sim a, b)$	1.59%	3.48%	4.22%
$\vec{\beta} = (a, b, \sim b)$	1.87%	3.56%	4.04%



Conclusion

- Catalytic power increases with dimension.
- Effective catalysts are always partially entangled; optimal structure is genuinely three-dimensional.
- The results offer an empirical path toward a dimension-dependent definition of catalytic power.

Next steps

- Extend simulations for catalysts beyond three dimensions.
- Rethink how faithfully Dirichlet-sampled probability vectors represent physically realizable quantum states.
- Search for analytic patterns distinguishing catalyzable pairs.

Technical Vocabulary

- Superposition:** an intrinsic property of quantum states; a state can exist as a combination of multiple possible outcomes at once, until measured.
- Entanglement:** a strong correlation between two quantum systems, such that the composite state cannot be written as a product of the individual systems' states.
- Pure bipartite quantum state:** a composite quantum system made of two distinct subsystems (labeled A and B), completely described by a single state vector $|\psi\rangle_{AB}$.
- Schmidt decomposition:** a way of expressing any pure bipartite state $|\psi\rangle_{AB}$ as a weighted sum over matched pairs of states from A and B.
- Local Operations and Classical Communication (LOCC):** a class of operations in which both A and B may freely act on their own system and communicate with each other classically but may not exchange quantum information.
- Majorization:** a mathematical relation that ranks how "spread out" or "concentrated" a probability distribution is.

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